MCA Semester 1 Previous Year Question Paper (2025)

University Examination: Data Structures & Advanced Database Systems

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SECTION A: Answer any 5 questions 5 x 4 = 20 marks.

Q1. Explain the time complexity of Quick Sort best, average, and worst cases.

Q2. Differentiate between B-Tree and B+ Tree indexing.

Q3. Explain ACID properties with a banking transfer example.

SECTION B: Answer any 3 questions 3 x 10 = 30 marks.

Q4. Construct an AVL tree by inserting keys: 15, 20, 24, 10, 13, 7, 30, 36, 25.

Q5. Discuss Conflict Serializability and precedence graph algorithm with diagram.